

Interaction-induced d_{xy} pairing in the checkerboard Hubbard model

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We compute the pair-field vertex of the half-filled Hubbard model on the checkerboard lattice by constrained-path quantum Monte Carlo on lattices of 8×8 , 10×10 and 12×12 sites. Separating the interaction-induced vertex from the uncorrelated part of the correlator, we find that the vertex vanishes identically in all four channels at $U = 0$, so every feature reported here is generated by the interaction. The on-site s vertex is negative throughout, as a repulsive interaction requires. The leading sign-changing channel changes with the lattice anisotropy δ , from extended s at small δ to d_{xy} above a crossover δ_c . That crossover is nearly independent of the interaction strength and its spread across U narrows from 0.148 to 0.028 between $L = 8$ and $L = 12$, converging on $\delta_c \simeq 0.19$, which identifies it as a property of the lattice geometry. The d_{xy} vertex grows with system size by a factor between 2.18 and 2.38 over the same range, and $d_{x^2-y^2}$ is positive in all 126 cells examined. The dominant channel lies outside the pairing basis used in earlier work on this lattice.

I. INTRODUCTION

Altermagnets combine a vanishing net moment with a spin-split band structure, a combination long assumed to require either ferromagnetism or spin-orbit coupling [1, 2]. The splitting follows from the crystal symmetry: the two opposite-spin sublattices are connected by a rotation rather than by a translation or an inversion, so the spin degeneracy that a conventional antiferromagnet enforces is absent.

Most microscopic models of this state build the splitting into the Hamiltonian, usually through a spin-dependent hopping. The splitting is then present already at zero interaction and the question of whether interactions alone can produce it is left open. Two recent studies of the Lieb lattice state the gap directly, noting that unbiased studies establishing an altermagnet as the ground state of an interacting Hamiltonian are lacking [3, 4]. Both use methods with a controlled range of validity, unrestricted Hartree-Fock with exact diagonalization on small clusters in the first case and a functional renormalization group treatment of the weak-coupling limit in the second.

The checkerboard lattice offers a route to the same physics without putting it in by hand. It is a square lattice with diagonal bonds on alternating plaquettes, so the two sublattices are exchanged by a fourfold rotation about a plaquette center but by no translation. The hopping is spin independent, and the interaction is a uniform on-site Hubbard U . Any spin splitting therefore appears only once magnetic correlations develop, which is the emergent route rather than the built-in one.

We report constrained-path quantum Monte Carlo (CPQMC) calculations of the pair-field vertex for this model at half filling on lattices of 8×8 , 10×10 and 12×12 sites. The vertex is the part of the pair correlator that the interaction builds, and it is the quantity that distinguishes a pairing tendency from the large positive correlator that free electrons already carry. Three results follow. The vertex is zero in every channel at $U = 0$. The

leading sign-changing channel is d_{xy} above an anisotropy δ_c whose value converges with system size. The d_{xy} vertex grows with system size at every interaction strength studied, while $d_{x^2-y^2}$ remains positive throughout and is never repulsive.

II. MODEL AND METHOD

We consider

$$H = \sum_{ij\sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (1)$$

with nearest-neighbor hopping $t_{ij} = t_0$ and diagonal hopping $t_1 + t_2$ along $(1, 1)$ and $t_1 - t_2$ along $(1, -1)$ on one sublattice, the two exchanged on the other. Energies are in units of $|t_0|$, and we set $t_0 = -1$ and $t_1 = 0.3$ throughout. The anisotropy is parametrized by δ through $t_2 = -\delta$, so that $\delta = 0$ restores the plain square lattice with uniform second-neighbor hopping and both sublattices become equivalent. All calculations are at half filling.

The hopping in Eq. (1) is spin independent by construction, so $n_\uparrow(\mathbf{k}) = n_\downarrow(\mathbf{k})$ at $U = 0$ for every δ .

We use CPQMC with a Trotter step $\Delta\tau = 0.01$, a projection length $\beta = 32$, and $N_w = 1000$ walkers. The constrained-path approximation introduces a bias controlled by the trial wave function, so we do not describe these results as exact. Doubling the projection length from $\beta = 32$ to $\beta = 48$ changes the order parameter by between 0.1% and 4.1% with no systematic drift, which we take as convergence in β .

The pair-field correlator is measured in four channels, on-site s , extended s , $d_{x^2-y^2}$ and d_{xy} , each defined by a form factor f_δ on the bonds it connects. The extended s and $d_{x^2-y^2}$ channels use the four nearest-neighbor bonds with f uniform and with $f = \pm 1$ on the x and y bonds respectively, and the d_{xy} channel uses the four diagonal bonds with $f = \pm 1$ on the two diagonal orientations. We report the interaction-induced vertex, which is the

correlator with its uncorrelated Wick contribution subtracted. The subtraction matters because the full correlator is large and positive already for free electrons, so only the vertex distinguishes a pairing tendency built by the interaction from the kinematic background.

III. PAIRING IN THE SAME REPRESENTATION

We compute the pair-field susceptibility in four channels, on-site s , extended s , $d_{x^2-y^2}$, and d_{xy} , and separate the interaction-induced vertex from the uncorrelated part. The vertex is the meaningful measure of whether the interaction builds pairing, since the full correlator is large and positive already for free electrons.

Three features stand out. The vertex is zero in all four channels at $U = 0$, so everything that follows is interaction generated. The on-site s vertex is negative throughout, as a repulsive U requires. The d_{xy} vertex is the largest of the four for $\delta \geq 0.3$ at every U studied, and it grows monotonically with U , from 3.347(8) at $U = 2$ to 5.952(22) at $U = 4.5$, in both cases peaking at $\delta = 0.4$.

The leading channel changes with anisotropy. For $\delta \leq 0.2$ the extended s channel is largest, and for $\delta \geq 0.3$ the d_{xy} channel is. This crossover falls in the same range of δ where the slope of Δ_{tot} with respect to U changes sign, so two quantities measured independently change character together.

The d_{xy} channel therefore leads over the whole range of anisotropy in which the vertex is appreciable, and it does so at every interaction strength studied.

IV. SIZE DEPENDENCE OF THE PAIRING VERTEX

The pairing vertex of Sec. III was computed at $L = 8, 10$, and 12 under identical settings, which allows the anisotropy of the channel crossover to be followed with size.

The crossover anisotropy δ_c , defined as the value at which the d_{xy} vertex overtakes the extended s vertex, is nearly independent of U and becomes more so as the lattice grows. Its spread across the six interaction strengths studied is 0.148 at $L = 8$, 0.049 at $L = 10$, and 0.028 at $L = 12$, and the mean value moves from 0.235 to 0.197 to 0.192. A crossover that is set by the interaction strength would not converge in this way, so we read $\delta_c \approx 0.19$ as a property of the lattice geometry.

The d_{xy} vertex grows with system size at every interaction strength, by a factor between 2.18 and 2.38 from $L = 8$ to $L = 12$. The $d_{x^2-y^2}$ vertex is positive in all 126 cells at all three sizes, so that channel is never repulsive even where it is subdominant. The optimal anisotropy for d_{xy} is $\delta = 0.4$ at essentially every size and interaction strength.

The two d channels respond with opposite sign to the same change in anisotropy at fixed interaction, so they are not simply rising and falling together with the overall pairing scale.

V. ROBUSTNESS AND COMPARISON

At half filling with periodic boundaries the noninteracting gap of Eq. (1) is zero at every L and δ we examined, so the occupied set is fixed by the trial state rather than by the spectrum. We repeated the calculation with antiperiodic boundaries, which shift the momentum mesh off the high-symmetry points and open a gap, at otherwise identical settings.

The closest published work is a determinant quantum Monte Carlo study of the checkerboard Hubbard model that reports d -wave and $d+is$ pairing as the dominant channels [5]. The two Hamiltonians are not the same. That model keeps one diagonal per sublattice and omits the other, whereas Eq. (1) keeps both, and the two coincide only where our second diagonal vanishes, at $\delta = t_1 = 0.3$. We verified that the single-particle spectra agree there to within 4×10^{-15} .

At that point, and at half filling, we evaluated the vertex in the basis of Ref. [5] alongside our own. Within their basis the s -wave channel is the larger of the two at $U = 3, 4$, and 4.5 . Including d_{xy} , which their basis does not contain, gives a d_{xy} vertex larger than either. The comparison is restricted, since Ref. [5] works at density $n = 0.9$ and temperature $T/t = 1/6$ while our calculations are at half filling in the ground state, and under the particle-hole transformation that relates the two signs of the second-neighbor hopping their density maps to $n = 1.1$. The statement is that the dominant channel we find lies outside the basis used there, not that the two calculations disagree.

VI. CONCLUSION

The interaction-induced pairing vertex of the half-filled checkerboard Hubbard model is zero in every channel at $U = 0$ and is largest in the d_{xy} channel above an anisotropy $\delta_c \simeq 0.19$. The crossover is nearly independent of the interaction strength and converges with system size, which identifies it with the lattice geometry rather than with the coupling. The d_{xy} vertex grows with system size at every interaction strength studied, and $d_{x^2-y^2}$ is positive in all 126 cells examined and therefore never repulsive.

The dominant channel lies outside the basis used in the closest published calculation on this lattice, so the comparison is one of basis rather than of disagreement. Whether the d_{xy} tendency survives to the thermodynamic limit is not settled by the sizes reached here, and the constrained-path bias remains.

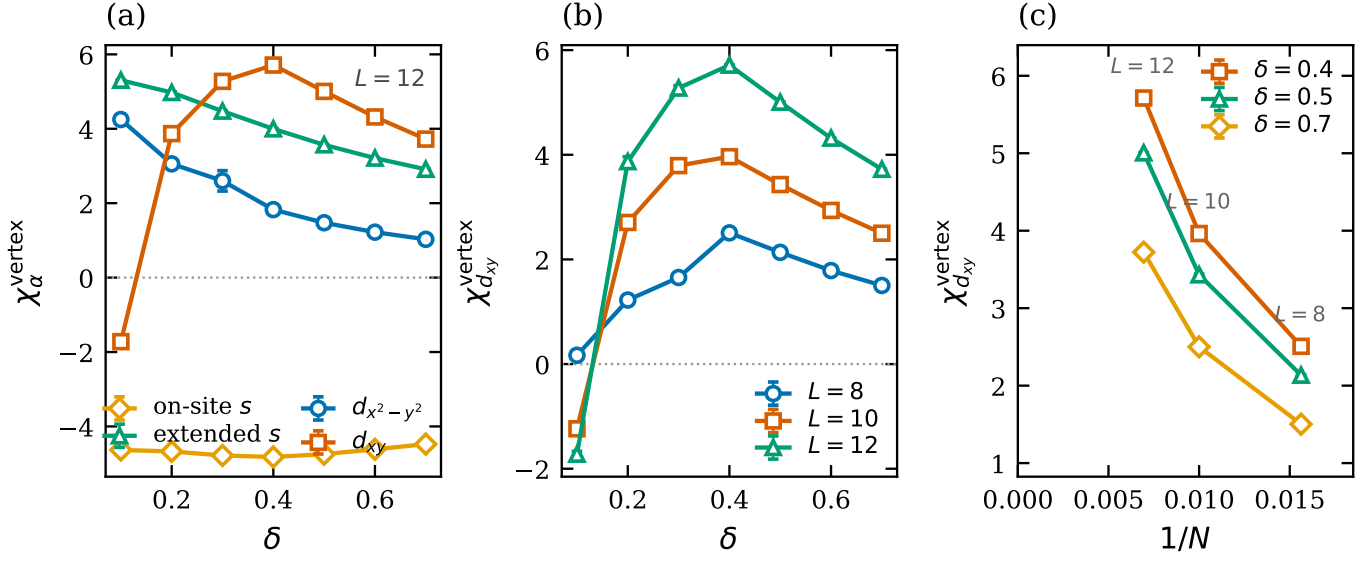


FIG. 1. Time-integrated pairing vertex at half filling. (a) the four channels at $L = 12$ and $U = 4$, (b) the d_{xy} vertex against δ for $L = 8, 10$ and 12 , and (c) the same quantity against $1/N$. The vertex is the interaction-induced part of the pair-field correlator and vanishes identically at $U = 0$ in all four channels. d_{xy} grows with system size at every interaction strength studied.

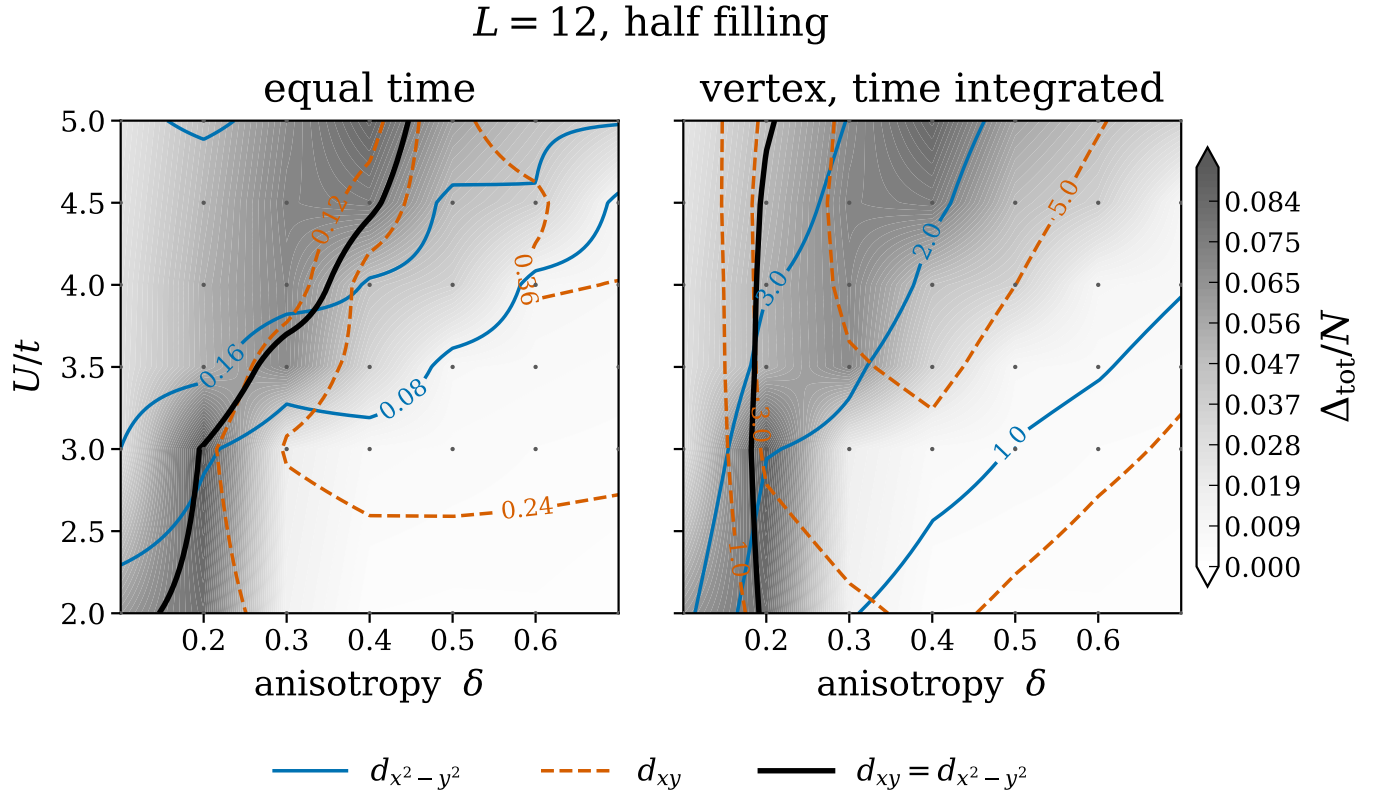


FIG. 2. The two d channels measured equal time (left) and time integrated (right) at $L = 12$. The equal-time correlator carries a large uncorrelated part, so the vertex is the measure used for the claims in the text.

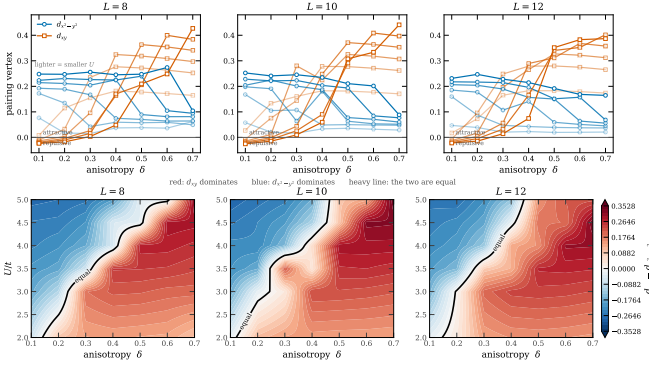


FIG. 3. Both d channels in the (U, δ) plane at $L = 12$. $d_{x^2-y^2}$ is positive everywhere, so it is never repulsive even where it is subdominant, and d_{xy} is the larger of the two above $\delta \simeq 0.19$.

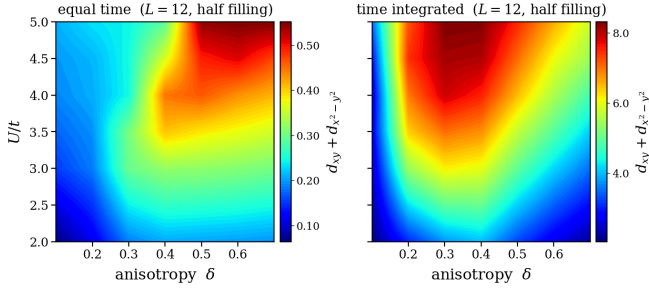


FIG. 4. Sum of the d_{xy} and $d_{x^2-y^2}$ vertices at $L = 12$, showing the total weight carried by the two sign-changing channels.

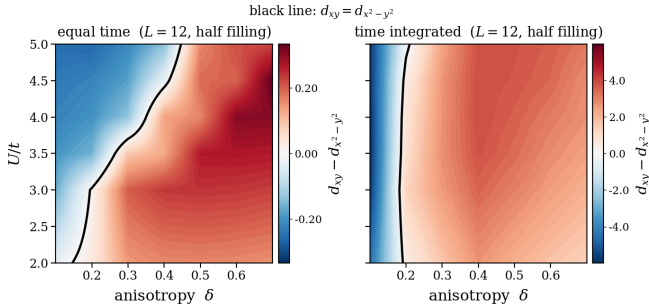


FIG. 5. Difference of the d_{xy} and $d_{x^2-y^2}$ vertices at $L = 12$, on the same colour scale as the preceding figure.

Appendix A: Trial wave function and the selection of the constrained path

The constrained-path approximation replaces the sign problem with a bias fixed by the trial wave function, so the trial is part of the method rather than an implementation detail. We take the trial from an unrestricted Hartree-Fock solution of Eq. (1) at the same U , δ , and filling, and we select among the solutions reached from different starting densities by the variational energy

$$E_{\text{HF}} = \text{Tr}(K\rho_{\uparrow}) + \text{Tr}(K\rho_{\downarrow}) + U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle, \quad (\text{A1})$$

with $\rho_{\sigma} = V_{\sigma} V_{\sigma}^{\dagger}$ built from the occupied columns of the mean-field eigenvectors and K the one-body matrix of Eq. (1).

Equation (A1) is not interchangeable with the sum of mean-field eigenvalues. Each eigenvalue already contains the mean field of the opposite spin, so that sum counts the interaction twice and orders competing solutions differently from the energy. The two criteria can select different states, and because the constrained path is fixed by whichever state is chosen, the selection propagates into every measured quantity.

The size of the effect is not small. Selecting by the eigenvalue sum instead of Eq. (A1) changes the trial in a substantial fraction of parameter sets, and at half filling the state chosen that way carries the larger staggered moment in every case where the two differ. We therefore report results obtained with Eq. (A1) throughout.

The identity $E_{\text{HF}} = \sum_p \epsilon_p - U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle$ holds only at exact self consistency and should not be substituted for Eq. (A1). A mean-field iteration that mixes densities between steps does not reach that condition for loosely converged solutions, which are precisely the ones whose ranking is at issue.

Appendix B: Convergence in projection length

All results use a projection length $\beta = 32$ in units of $1/|t_0|$. Doubling it to $\beta = 48$ at otherwise identical settings changes Δ_{tot} by between 0.1% and 4.1% with no systematic drift in either direction, which we take as convergence.

Appendix C: Boundary conditions

At half filling with periodic boundaries the noninteracting spectrum of Eq. (1) is gapless at every L and δ examined, so the occupied set is fixed by the trial state rather than by the spectrum. Antiperiodic boundaries shift the momentum mesh off the high-symmetry points and open a gap.

The sensitivity of the calculation to the momentum mesh was characterized on the magnetic response rather

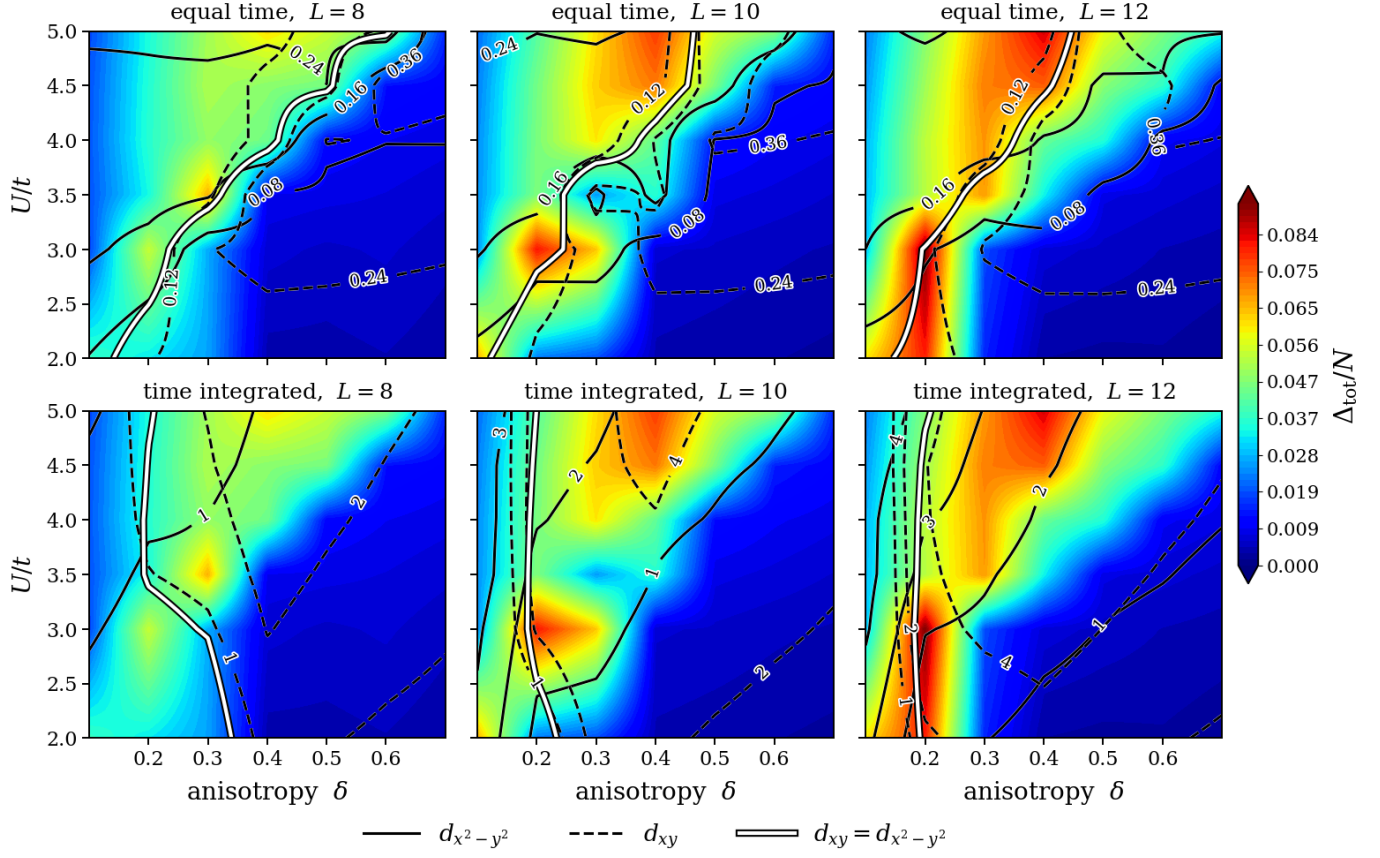


FIG. 6. Equal-time pair correlator (top row) and time-integrated vertex (bottom row) for $L = 8, 10$ and 12 . The channel crossover straightens and its spread across U narrows along the vertex row, which is the behavior expected of a quantity fixed by the lattice geometry rather than by the coupling.

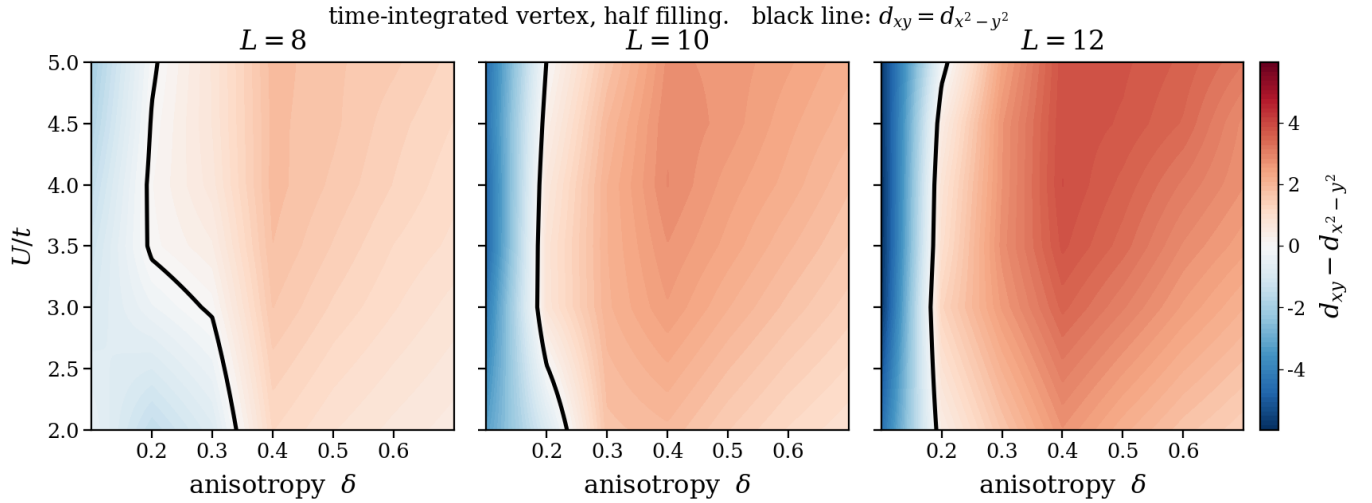


FIG. 7. Difference between the d_{xy} and $d_{x^2-y^2}$ vertices over the (U, δ) plane at $L = 8, 10$ and 12 . The zero contour is the crossover anisotropy δ_c , which is nearly flat in U and whose spread across the interaction strengths studied falls from 0.148 to 0.049 to 0.028 as the lattice grows.

than on the vertex. Its slope in U agrees between the two boundary conditions at $\delta = 0.1$ and at $\delta = 0.5$ and is ambiguous at $\delta = 0.3$, where that slope passes through zero and its sign is most sensitive to the mesh. We have not repeated the vertex calculation with antiperiodic boundaries, so the mesh dependence of the pairing results is

not quantified here.

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